

UAV-Based Efficient Joint Estimation of Directional Emitter Position and Transmission Orientation

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Abstract—This letter addresses the problem of unmanned aerial vehicles (UAVs)-based localization for a directional emitter with unknown position and transmission orientation. We propose an efficient three-step alternating estimator with iteratively reweighted least square (TA-IRLS). Based on linearized measurement models, the initial estimation of emitter position and orientation are obtained using low-complexity least squares (LS) approaches. To further enhance localization accuracy, IRLS is employed to refine measurement weights based on estimation residuals. Numerical results validate the superiority of the proposed TA-IRLS in terms of both localization efficiency and accuracy.

Index Terms—Wireless signal localization, directional emitter, RSS, AOA, IRLS.

I. INTRODUCTION

PRECISE position information of radio frequency (RF) signal emitters is crucial for various applications, including interference localization and search and rescue. Recently, unmanned aerial vehicle (UAV)-assisted signal localization has demonstrated significant potential thanks to flexibility and cost-effectiveness [1], [2], [3].

Signal emitter positions can be estimated using various types of signal measurements, such as angle of arrival (AOA), received signal strength (RSS), time of arrival (TOA), time difference of arrival (TDOA), or a combination of these measurements [4], [5], [6]. Wherein, TOA- and TDOA-based algorithms require perfect time synchronization, which is challenging in UAV-assisted localization due to high communication delay. RSS-based localization algorithms, as explored by Gao et al. through an alternating minimization algorithm [4], provide a practical solution. However, RSS alone may have limitations in terms of localization accuracy. Therefore, combining RSS with AOA measurements have garnered considerable attention [7], [8]. The authors

in [8] presented a nonconvex estimator utilizing RSS and AOA measurements based on the least squares criterion and solved it by semi-definite programming (SDP) relaxation. Alternatively, second-order cone programming (SOCP) was proposed in [9] for a more efficient solution compared to SDP. Khan et al. introduced a linear least squares (LS) method that is straightforward to implement with linear computational complexity [10]. Furthermore, Tomic et al. linearized the measurement models of RSS and AOA by exploiting spherical coordinate conversion and derived a weighted least squares (WLS) approach which improves efficiency and accuracy [11].

Despite these advancements, there has been limited focus on directional emitters, which are increasingly prevalent in emerging applications such as satellite communication [12]. The maximum likelihood estimation of both position and orientation for such emitters requires high-dimensional grid searches, which are computationally extensive. Zuo et al. addressed this challenge by employing second-order cone relaxation for emitter position estimation and a multi-resolution (MR) search technique for transmission orientation estimation [13]. In a subsequent study [5], they reformulated the problem of RSS-TOA-based localization as a generalized trust region subproblem (GTRS) and solved it via a bisection process, followed by the application of the MR search technique.

Nevertheless, the limited computational capabilities of the UAVs necessitate algorithms that can provide rapid solutions with low complexity while maintaining high accuracy. To address this demand, this letter proposes a three-step alternating estimator based on iteratively reweighted least squares (TA-IRLS) method. The measurement models are linearized to facilitate the initial estimation of position and transmission orientation using LS approaches, effectively reducing computational complexity. Subsequently, the localization accuracy is further refined through IRLS, where weights are adaptively adjusted based on the estimation residuals. Compared to WLS methods that employ static weights, this dynamic adjustment allows the algorithm to mitigate the effects of measurements with high errors, thereby enhancing localization accuracy.

II. SYSTEM MODEL AND PROBLEM FORMULATION

As depicted in Fig. 1, we consider a localization system where M UAVs are employed to localize an unknown directional RF emitter. The coordinates and transmission orientation of the emitter can be denoted as $\mathbf{x} = [x_1, x_2, x_3]^T$ and $\boldsymbol{\eta} = [\phi_b, \theta_b]^T$, respectively. Without loss of generality, we assume that the UAVs' positions $\mathbf{a}_i = [a_{i1}, a_{i2}, a_{i3}]^T$, $i = 1, \dots, M$ can be accurately acquired. The UAVs are equipped with antenna arrays so that the RSS and AOA

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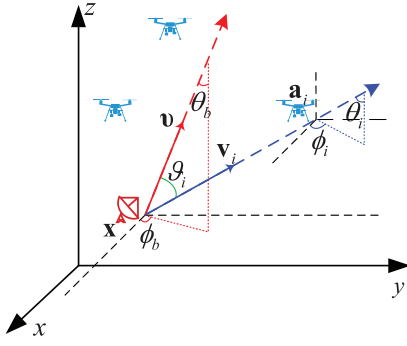


Fig. 1. The scenario of jointly estimating directional emitter position and transmission orientation.

measurements can be obtained from the signals transmitted by the RF emitter. It is assumed that the emitter is equipped with a directional antenna with Gaussian-shaped radiation pattern $|G(\vartheta_i)|^2 = \exp(-\frac{\vartheta_i^2}{2\beta^2})$ [14], where β represents the half-power beamwidth; ϑ_i represents the off-axis angle, which is defined as the angle between the unit vector $\mathbf{v}$ of transmission orientation and the unit vector $\mathbf{v}_i$ pointed from emitter to the i -th UAV. The off-axis angle can be obtained by $\vartheta_i = \cos^{-1}(\frac{\mathbf{v}^T \mathbf{v}_i}{\|\mathbf{v}\| \|\mathbf{v}_i\|})$, where $\mathbf{v} = [\sin(\theta_b) \cos(\phi_b), \sin(\theta_b) \sin(\phi_b), \cos(\theta_b)]^T$; $\mathbf{v}_i = \frac{\mathbf{a}_i - \mathbf{x}}{\|\mathbf{a}_i - \mathbf{x}\|} = [\frac{a_{i1} - x_1}{d_i}, \frac{a_{i2} - x_2}{d_i}, \frac{a_{i3} - x_3}{d_i}]^T$.

The AOA measurements obtained by the i -th UAV are expressed as

$$\phi_i = \tan^{-1}\left(\frac{x_2 - a_{i2}}{x_1 - a_{i1}}\right) + n_{\phi_i}, \quad (1a)$$

$$\theta_i = \cos^{-1}\left(\frac{x_3 - a_{i3}}{\|\mathbf{x} - \mathbf{a}_i\|}\right) + n_{\theta_i}, \quad (1b)$$

where $n_{\phi_i} \sim \mathcal{N}(0, \sigma_{\phi_i}^2)$ and $n_{\theta_i} \sim \mathcal{N}(0, \sigma_{\theta_i}^2)$ represent the measurement noises of azimuth angle and elevation angle, respectively. The noise-free RSS measured by the i -th UAV can be expressed as $P_i = P_0 (\frac{\|\mathbf{x} - \mathbf{a}_i\|}{d_0})^{-\gamma} \exp(-\frac{\vartheta_i^2}{2\beta^2})$, where P_0 denoted the RSS at a reference distance d_0 . The RSS in decibels (dBm) is described as

$$P_i = P_0 - 10\gamma \log_{10}\left(\frac{\|\mathbf{x} - \mathbf{a}_i\|}{d_0}\right) - \frac{5\vartheta_i^2}{\beta^2 \ln 10} + n_{P_i}, \quad (2)$$

where n_{P_i} denotes the log-normal shadowing term (dB) modeled as $n_{P_i} \sim \mathcal{N}(0, \sigma_{P_i}^2)$; γ denotes path loss exponent (PLE).

To summarize, the observation vector can be given by $\mathbf{O} = [\mathbf{P}^T, \boldsymbol{\phi}^T, \boldsymbol{\theta}^T]^T \in \mathbb{R}^{3M \times 1}$, where $\mathbf{P}^T = [P_1, \dots, P_M]^T$, $\boldsymbol{\phi}^T = [\phi_1, \dots, \phi_M]^T$, $\boldsymbol{\theta}^T = [\theta_1, \dots, \theta_M]^T$. The noise vector can be represented as $\boldsymbol{\sigma} = [\sigma_{P_1}, \dots, \sigma_{P_M}, \sigma_{\phi_1}, \dots, \sigma_{\phi_M}, \sigma_{\theta_1}, \dots, \sigma_{\theta_M}]^T$. Let us define $\boldsymbol{\xi} = [\mathbf{x}^T, \boldsymbol{\eta}^T]^T \in \mathbb{R}^{5 \times 1}$ and $\mathbf{f}(\boldsymbol{\xi}) = [P_0 - 10\gamma \log_{10}(\frac{\|\mathbf{x} - \mathbf{a}_i\|}{d_0}) - \frac{5\vartheta_i^2}{\beta^2 \ln 10}, \tan^{-1}(\frac{x_2 - a_{i2}}{x_1 - a_{i1}}), \cos^{-1}(\frac{x_3 - a_{i3}}{\|\mathbf{x} - \mathbf{a}_i\|})]^T$. The conditional probability density function, which quantifies the joint likelihood of the observed AOA and RSS measurements given the state $\boldsymbol{\xi}$, can be expressed as

$$p(\mathbf{O}|\boldsymbol{\xi}) = \prod_{m=1}^{3M} \frac{1}{\sqrt{2\pi\sigma_m^2}} \exp\left\{-\frac{[O_m - f_m(\boldsymbol{\xi})]^2}{2\sigma_m^2}\right\}. \quad (3)$$

Therefore, the maximum likelihood estimation (MLE) of the emitter position and transmission orientation can be formulated by

$$\hat{\boldsymbol{\xi}} = \arg \min_{\boldsymbol{\xi}} \sum_{m=1}^{3M} \frac{1}{\sigma_m^2} [O_m - f_m(\boldsymbol{\xi})]^2. \quad (4)$$

III. PROPOSED ESTIMATOR

The formulated MLE in (4) is highly non-convex. Moreover, directly solving it requires a 5-D grid search, which is extremely computationally expensive. In this section, we introduce an efficient estimator TA-IRLS to solve the problem.

To streamline the analysis, we adopt the assumption that the measurement noises are sufficiently small, as commonly done in previous studies [8]. The following approximations can be derived based on (1a), (1b), and (2), respectively [13]:

$$\mathbf{c}_i^T (\mathbf{x} - \mathbf{a}_i) \approx 0, \quad (5a)$$

$$\mathbf{q}^T (\mathbf{x} - \mathbf{a}_i) \approx \|\mathbf{x} - \mathbf{a}_i\| \cos \theta_i, \quad (5b)$$

$$\lambda_i \|\mathbf{x} - \mathbf{a}_i\| \approx \mu_i d_0, \quad (5c)$$

where $\mathbf{c}_i = [-\sin(\phi_i), \cos(\phi_i), 0]^T$, $\mathbf{q} = [0, 0, 1]^T$, $\lambda_i = 10^{\frac{P_i - P_0}{-10\gamma}}$, $\mu_i = 10^{-\frac{\delta \vartheta_i^2}{10\gamma}}$, $\delta = \frac{5}{\beta^2 \ln 10}$. By resorting to spherical coordinates [11], we first rewrite the norm term as $\|\mathbf{x} - \mathbf{a}_i\| = \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i)$ where $\mathbf{u}_i = [\cos \phi_i \sin \theta_i, \sin \phi_i \sin \theta_i, \cos \theta_i]^T$ with $\|\mathbf{u}_i\| = 1$. This conversion leverages the properties of spherical space to linearize the measurement models, making the estimation problem easier to solve. Then by applying the least squares (LS) approach, the estimator (4) can be rewritten as

$$\begin{aligned} \hat{\boldsymbol{\xi}} = \arg \min_{\boldsymbol{\xi}} & \sum_{i=1}^M (\mathbf{c}_i^T (\mathbf{x} - \mathbf{a}_i))^2 \\ & + \sum_{i=1}^M (\mathbf{q}^T (\mathbf{x} - \mathbf{a}_i) - \mu_i d_0 \cos \theta_i)^2 \\ & + \sum_{i=1}^M (\lambda_i \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i) - \mu_i d_0)^2. \end{aligned} \quad (6)$$

It is obvious that the first and second terms (which correspond to AOA measurements) of the right side of (6) are not related to the transmission orientation $\boldsymbol{\eta}$. Moreover, the third term (which correspond to the RSS measurements) is related to both the emitter position $\mathbf{x}$ and the transmission orientation $\boldsymbol{\eta}$. Consequently, $\mathbf{x}$ and $\boldsymbol{\eta}$ can be estimated alternatively [13]. Define $F_1 = \sum_{i=1}^M (\mathbf{c}_i^T (\mathbf{x} - \mathbf{a}_i))^2 + \sum_{i=1}^M (\mathbf{q}^T (\mathbf{x} - \mathbf{a}_i) - \mu_i d_0 \cos \theta_i)^2$ and $F_2 = \sum_{i=1}^M (\lambda_i \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i) - \mu_i d_0)^2$. The problem (6) can be reorganized as $\hat{\boldsymbol{\xi}} = \arg \min_{\boldsymbol{\xi}} F_1 + F_2$.

To start with, the emitter position can be estimated by $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} F_1$, which can be rewritten as $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{A}_1 \mathbf{x} - \mathbf{b}_1\|^2$, where $\mathbf{A}_1 = [\dots, \mathbf{c}_m^T, \dots, (\cos \theta_m \mathbf{u}_m - \mathbf{q})^T, \dots]^T$, $\mathbf{b}_1 = [\dots, \mathbf{c}_m^T \mathbf{a}_m, \dots, (\cos \theta_m \mathbf{u}_m - \mathbf{q})^T \mathbf{a}_m, \dots]^T$. The closed-form solution can be obtained by

$$\hat{\mathbf{x}} = (\mathbf{A}_1^T \mathbf{A}_1)^{-1} \mathbf{A}_1^T \mathbf{b}_1, \quad (7)$$

Subsequently, we substitute the estimated emitter position $\hat{\mathbf{x}}$ into F_2 to estimate the auxiliary variable $\boldsymbol{\mu} = [\mu_1, \dots, \mu_M]^T$, i.e., $\hat{\boldsymbol{\mu}} = \arg \min_{\boldsymbol{\mu}} F_2$, which can be obtained by

$$\hat{\mu}_i = \lambda_i \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i) / d_0. \quad (8)$$

Note that $\frac{\mu_i}{\delta} = 10^{-\frac{\delta \vartheta_i^2}{10\gamma}}$, we can obtain $\hat{\vartheta}_i = \sqrt{-10\gamma \log_{10} \hat{\mu}_i / \delta}$ for $i = 1, \dots, M$.

Based on $\hat{\mathbf{x}}$, we can then calculate $\hat{\mathbf{v}}_i = \frac{\mathbf{a}_i - \hat{\mathbf{x}}}{\|\mathbf{a}_i - \hat{\mathbf{x}}\|}$. The unit vector of transmission orientation $\mathbf{v}$ can be estimated from $\hat{\mathbf{v}}_i$ and $\hat{\vartheta}_i$, i.e., $\hat{\mathbf{v}} = \sum_{i=1}^M \arg \min_{\mathbf{v}} (\frac{\mathbf{v}^T \hat{\mathbf{v}}_i}{\|\mathbf{v}\| \cdot \|\hat{\mathbf{v}}_i\|} - \cos \hat{\vartheta}_i)$. This problem can be represented as $\hat{\mathbf{v}} = \min_{\mathbf{v}} \|\mathbf{A}_2 \mathbf{v} - \mathbf{b}_2\|^2$, where $\mathbf{A}_2 = [\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_M]^T$, $\mathbf{b}_2 = [\cos(\hat{\vartheta}_1), \dots, \cos(\hat{\vartheta}_M)]^T$. The closed-form solution can be given by

$$\hat{\mathbf{v}} = (\mathbf{A}_2^T \mathbf{A}_2)^{-1} \mathbf{A}_2^T \mathbf{b}_2. \quad (9)$$

Finally, the estimated transmission orientation $\boldsymbol{\eta} = [\phi_b, \theta_b]^T$ can be obtained by

$$\hat{\theta}_b = \cos^{-1}(\hat{\mathbf{v}}_{3,1}), \quad (10a)$$

$$\hat{\phi}_b = \tan^{-1} \left(\frac{\hat{\mathbf{v}}_{2,1} / \sin \hat{\theta}_b}{\hat{\mathbf{v}}_{1,1} / \sin \hat{\theta}_b} \right). \quad (10b)$$

To further refine localization accuracy, the estimated transmission orientation $\hat{\boldsymbol{\eta}}$ can be utilized to estimate the emitter position by using both AOA and RSS measurements, i.e., $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} F_1 + F_2$. Considering the varying importance of noisy measurements acquired by different UAVs, we initially formulate it as a weighted least square (WLS) problem:

$$\min_{\mathbf{x}} \|\mathbf{W}(\mathbf{A}_3 \mathbf{x} - \mathbf{b}_3)\|^2, \quad (11)$$

where $\mathbf{A}_3 = [\dots, \mathbf{c}_m^T, \dots, (\cos \theta_m \mathbf{u}_m - \mathbf{q})^T, \dots, \lambda_m \mathbf{u}_m^T, \dots]$, $\mathbf{b}_3 = [\dots, \mathbf{c}_m^T \mathbf{a}_m, \dots, (\cos \theta_m \mathbf{u}_m - \mathbf{q})^T \mathbf{a}_m, \dots, \lambda_m \mathbf{u}_m^T \mathbf{a}_m + \hat{\mu}_m d_0, \dots]$, $\mathbf{W}$ is the weight matrix. The WLS methods in [8], [11] use fixed weights determined by estimated distances to assign more confidence to sensors that is closer to the target. However, the sensors close to the emitter do not necessarily receive high-SNR measurements for reliable localization considering high directivity of signals. To further enhance robustness and accuracy of localization, we utilize an iteratively reweighted least squares (IRLS) [15] method, allowing adaptive adjustments to the weights based on the residuals from the previous iteration.

The estimated $\hat{\mathbf{x}}$ obtained from (7) is set as the initial estimation of emitter position $\hat{\mathbf{x}}^{(0)}$ in IRLS. For each iteration k , the weight matrix can be represented as $\mathbf{W}^{(k)} = \text{diag}(\dots, w_{i,1}^{(k)}, \dots, w_{i,2}^{(k)}, \dots, w_{i,3}^{(k)}, \dots) \in \mathbb{R}^{3M \times 3M}$, where $w_{i,1}^{(k)} = \frac{1}{\|r_{i,1}^{(k)}\| + \rho}$, $w_{i,2}^{(k)} = \frac{1}{\|r_{i,2}^{(k)}\| + \rho}$, $w_{i,3}^{(k)} = \frac{1}{\|r_{i,3}^{(k)}\| + \rho}$. In these equations, ρ is a small constant introduced to prevent division by zero. Moreover, the residuals can be calculated from the previous estimation $\hat{\mathbf{x}}^{(k-1)}$, that is, $r_{i,1}^{(k)} = \mathbf{c}_i^T (\hat{\mathbf{x}}^{(k-1)} - \mathbf{a}_i)$, $r_{i,2}^{(k)} = \mathbf{q}^T (\mathbf{x} - \mathbf{a}_i) - \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i) \cos \theta_i$, $r_{i,3}^{(k)} = \lambda_i \mathbf{u}_i^T (\mathbf{x} - \mathbf{a}_i) - \mu_i d_0$. Then the emitter position can be estimated by

Algorithm 1 The Proposed TA-IRLS Algorithm

Input: $\mathbf{a}_i, \phi_i, \theta_i$ for $i = 1, 2, \dots, M$.

Output: Estimated position $\hat{\mathbf{x}}$, and estimated transmission orientation $\hat{\theta}_b$ and $\hat{\phi}_b$.

- 1: Initialize $K = 1$;
****** Obtain initial estimation of $\hat{\mathbf{x}}^{(0)}$ ******
- 2: Estimate $\hat{\mathbf{x}}^{(0)}$ according to (7);
- 3: Calculate $\hat{\mathbf{v}}_i$ for $i = 1, 2, \dots, M$;
****** Estimate θ_b and ϕ_b ******
- 4: Estimate $\hat{\mu}_i$ for $i = 1, 2, \dots, M$ by (8);
- 5: Calculate off-axis angle $\hat{\vartheta}_i$ for $i = 1, 2, \dots, M$;
- 6: Estimate $\hat{\mathbf{v}}$ by (9);
- 7: Calculate estimated $\hat{\theta}_b$ and $\hat{\phi}_b$ by (10);
****** Refine the estimation of $\hat{\mathbf{x}}$ ******
- 8: **while** $k < K$ or $\|\hat{\mathbf{x}}^{(k)} - \hat{\mathbf{x}}^{(k-1)}\| > \epsilon$ **do**
- 9: Refine the estimation of $\hat{\mathbf{x}}^{(k)}$ iteratively by (12);
- 10: $k = k + 1$;
- 11: **end while**

TABLE I
COMPARISON OF COMPLEXITY AND RUN TIME

Methods	Complexity	Run time
Proposed	$\mathcal{O}(KM)$	0.004s
SOCP-MR	$\mathcal{O}(M^{3.5} + QM)$	0.658s
SDP-MR	$\mathcal{O}(M^{4.5} + QM)$	1.117s
Grid-search	$\mathcal{O}((P + Q)M)$	3.342s

$$\hat{\mathbf{x}}^{(k)} = (\mathbf{A}_3^T \mathbf{W}^{(k)} \mathbf{A}_3)^{-1} \mathbf{A}_3^T \mathbf{W}^{(k)} \mathbf{b}_3. \quad (12)$$

The iteration terminates when $\|\hat{\mathbf{x}}^{(k)} - \hat{\mathbf{x}}^{(k-1)}\| \leq \epsilon$ or the maximum number K of iterations is reached.

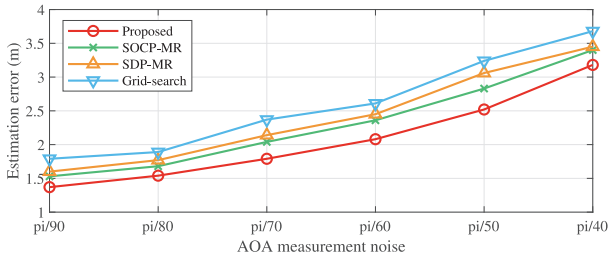
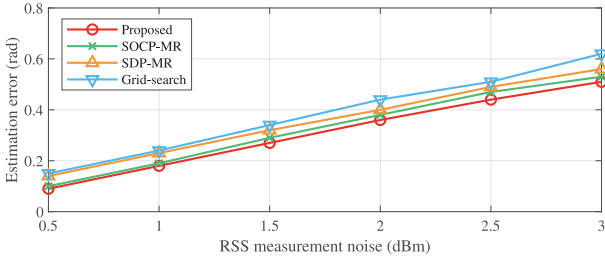
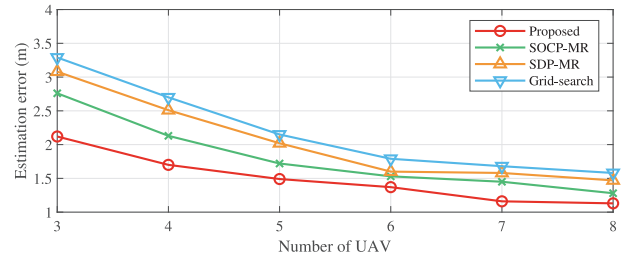
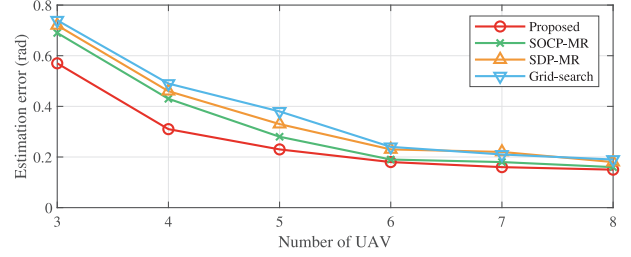
The procedures of the proposed TA-IRLS are summarized in Algorithm 1. In the first step, an initial estimation of $\hat{\mathbf{x}}^{(0)}$ is obtained utilizing merely AOA measurements. In the second step, the transmission orientation parameters $\hat{\theta}_b$ and $\hat{\phi}_b$ are estimated based on RSS measurements. Finally, the position estimation $\hat{\mathbf{x}}$ is iteratively refined by jointly leveraging both AOA and RSS measurements given $\hat{\theta}_b$ and $\hat{\phi}_b$, so as to refine localization accuracy by effectively integrating the information provided by these two types of measurements.

IV. COMPLEXITY ANALYSIS

The computational complexity of the proposed algorithm is analyzed as follows: both the first and second steps based on LS approaches have a complexity of $\mathcal{O}(M)$, while the third step has a worst-case complexity of $\mathcal{O}(KM)$, where K denotes the maximum number of iterations. Thus, the overall complexity is $\mathcal{O}(KM)$. In comparison, the computational complexity of SOCP-MR [13] is $\mathcal{O}(M^{3.5} + QM)$, where Q represents the times of grid search for orientation estimation. SDP has a complexity of $\mathcal{O}(M^{4.5})$ [9], resulting in $\mathcal{O}(M^{4.5} + QM)$ for SDP-MR. The complexity of grid-search [16] is $\mathcal{O}((P + Q)M)$, where P denotes the times of grid search for position estimation.

V. PERFORMANCE EVALUATION

In the given simulation scenario, the UAVs and directional emitter are randomly placed in a cube of $100\text{m} \times 100\text{m} \times 100\text{m}$

Fig. 2. Comparison of the RMSE of position estimation versus σ_ϕ , σ_θ .Fig. 3. Comparison of the RMSE of orientation estimation versus σ_P .Fig. 4. Comparison of the RMSE of position estimation versus M .Fig. 5. Comparison of the RMSE of orientation estimation versus M .

in each Monte Carlo trial. The transmission orientation of directional emitter is randomly set as well. In terms of signal propagation, we set $P_0 = 30\text{dBm}$, $d_0 = 1\text{m}$, $\gamma = 2.8$, $\beta = \pi/3$. The number of UAVs is $M = 6$, unless otherwise stated. For brevity, we assume that the measurement noise variances of each UAV are equivalent, i.e., $\sigma_{\phi_i} = \sigma_\phi = \frac{\pi}{90}$, $\sigma_{\theta_i} = \sigma_\theta = \frac{\pi}{90}$, $\sigma_{P_i} = \sigma_P = 1$ for $i = 1, \dots, M$, unless otherwise stated. Moreover, we set $\epsilon = 0.1$, $\rho = 10^{-4}$, $K = 100$.

To demonstrate the advantages of the proposed estimator, we compare it with following baselines: SOCP-MR [13]; SDP-MR which estimates $\mathbf{x}$ using SDP [8] and then follows the MR search adopted in [13] to estimate $\boldsymbol{\eta}$; Grid-search which directly estimates both $\mathbf{x}$ and $\boldsymbol{\eta}$. The searching step lengths of the 3-D coordinates are set as 2m, and the searching step lengths of azimuth and elevation angles are set as $\pi/60$. Consequently, the total times of grid search for position estimation and orientation estimation are $P = 125000$ and $Q = 7200$, respectively. The performance metrics include root mean square errors (RMSE) of emitter position estimation and transmission orientation estimation, which can be respectively represented as

$$\text{RMSE}_{\mathbf{x}} = \sqrt{\frac{1}{N} \sum_{n=1}^N \|\mathbf{x} - \hat{\mathbf{x}}_n\|^2}, \quad (13)$$

$$\text{RMSE}_{\boldsymbol{\eta}} = \sqrt{\frac{1}{N} \sum_{n=1}^N \|\cos^{-1}(\mathbf{v}^T \hat{\mathbf{v}}_n)\|^2}. \quad (14)$$

where $\hat{\mathbf{x}}_n$ and $\hat{\mathbf{v}}_n$ denote the estimated emitter position and transmission orientation in the n -th Monte Carlo trial. The performance is averaged over $N = 10000$ Monte Carlo trials.

We first evaluate the computational complexity and average running time of these algorithms. As shown in Table I, the grid-search method is the most time-consuming. Moreover, SDP-MR is also computationally intensive due to the high complexity of SDP. SOCP-MR achieves a reduction in running

time compared to SDP-MR, yet it remains less efficient than the proposed method. This can be accounted by the fact that the MR search method involves exhaustive searches over candidate orientations, which further contributes to the time consumption. It is noteworthy that our proposed estimator significantly reduces the running time. This superior performance can be attributed to the low complexity of LS.

Fig. 2 illustrates the RMSE of $\mathbf{x}$ versus the AOA measurement noise power. The RMSE exhibits an increasing trend as σ_ϕ and σ_θ increases. This is expected since increased noise in the angle measurements impairs AOA estimation, leading to less precise emitter localization. The proposed algorithm, benefiting from the enhanced estimation by utilizing hybrid measurements based on initial estimation of $\mathbf{x}$ and $\boldsymbol{\eta}$, achieves the lowest RMSE of $\mathbf{x}$. Moreover, our proposed algorithm is less prone to increases in noise power. This stems from adaptive adjustment of weights according to estimation residuals in the IRLS process. Similarly, Fig. 3 demonstrates that the RMSE of $\boldsymbol{\eta}$ increases with RSS measurement noise power σ_P . The underlying rationale is that the degradation in RSS measurement quality caused by higher noise levels adversely affects the estimation of transmission orientation. In addition, owing to the fact that the MR search method relies on the estimated emitter position, the position estimation errors in SOCP-MR and SDP-MR may further lead to deterioration of orientation estimation accuracy. The proposed algorithm outperforms benchmark algorithms, confirming the efficacy of IRLS which benefits from the effective linearization of the estimation problem.

Furthermore, we investigate the impact of the number M of UAVs on the RMSE of $\mathbf{x}$ and $\boldsymbol{\eta}$, as shown in Fig. 4 and Fig. 5, respectively. It can be observed that the RMSE achieved by all the algorithms decreases as M increases. This can be accounted for by the fact that additional measurements help to mitigate the influence of individual sensor noises, thereby enhancing the localization accuracy. Notably, the proposed algorithm consistently achieves the lowest RMSE among across all value

of M . The performance gain is particularly pronounced when M is small, highlighting the outstanding capability of our proposed algorithm even in scenarios with a limited number of UAVs.

VI. CONCLUSION

In this letter, we have investigated UAV-assisted joint estimation of position and transmission orientation for a directional emitter using hybrid RSS-AOA measurements. Considering the limited onboard computational capacity of UAVs, we have proposed an efficient estimator referred to as TA-IRLS, which takes advantages of the low complexity of LS approaches. Moreover, the localization accuracy is enhanced by an IRLS. Numerical results have demonstrated that the proposed method significantly outperforms existing techniques in terms of both localization efficiency and accuracy.

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